Squared & Cubic Units

Exam code: 0607

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Squared & cubic units

How do I convert between squared units (areas)?

- You need to **square** the **unit conversion** rates
  - E.g., 1 cm^2 = 10^2 mm^2 = 100 mm^2
  - This is because area is 2D
  - The fact the units have a 'squared' on them will help you remember
- It can help to **imagine a square**
  - E.g. 1 m^2 is a square measuring 1 m × 1 m
  - In cm this would be 100 cm × 100 cm
  - So 1 m^2 is equivalent to 10 000 cm^2
- The **basic conversions** for **area** are
  - 1 cm^2 = 100 mm^2
  - 1 m^2 = 10 000 cm^2
  - 1 km^2 = 1 000 000 m^2
- You may be told conversions for other units in a question, such as
  - 1 hectare (ha) = 10 000 m^2

How do I convert between cubed units (volume)?

- You need to **cube** the **unit conversion** rates
  - E.g. 1 cm^3 = 10^3 mm^3 = 1000 mm^3
  - This is because volume is 3D
  - The fact the units have a "cubed" on them will help you remember
- It can help to **imagine a cube**
  - E.g. 1 m^3 is a cube measuring 1 m × 1 m × 1 m
  - In cm this would be 100 cm × 100 cm × 100 cm
  - So 1 m^3 is equivalent to 1 000 000 cm^3
- The **basic conversions** for volume are
  - 1 cm^3 = 1000 mm^3
  - 1 m^3 = 1 000 000 cm^3
  - 1 km^3 = 1 000 000 000 m^3

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Convert

(a) 8254 mm^2 to cm^2

Answer:

1 cm = 10 mm

1 cm^2 = 10^2 mm^2 = 100 mm^2

8254 mm^2 = (8254 ÷ 100) cm^2

82.54 cm^2

(b) 2.54 m^3 to cm^3

Answer:

1 m = 100 cm
1 m^3 = 100^3 cm^3 = 1 000 000 cm^3

2.54 m^3 = (2.54 × 1 000 000) cm^3

2 540 000 cm^3

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